

SATTAHIP, Thailand

WMO: 484770

Lat: 12.6932N

Lon: 100.9820E

Elev: 21

StdP: 101.07

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	20.6	11.7	8.6	24.2	13.6	9.7	24.8	6.0	26.9	5.4	28.0	2.2	0	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	6.2	34.6	26.8	34.0	26.9	33.5	26.7	28.7	32.4	28.3	31.9	28.0	31.6	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.8	23.9	31.1	27.2	23.1	30.6	27.1	22.8	30.5	93.3	32.6	91.5	32.2	90.1	31.6	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.0	5.4	5.1	DB	17.0	37.1	1.8	1.4	15.8	38.1	14.7	38.9	13.7	39.7	12.5	40.7	
			WB	13.9	29.8	1.8	1.0	12.6	30.5	11.6	31.1	10.6	31.6	9.3	32.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	26.8	27.8	29.0	30.2	30.1	29.7	29.4	29.4	28.7	28.3	28.0	27.0
	DBStd	1.67	1.48	1.37	1.37	1.32	1.28	1.09	1.14	0.98	1.08	0.95	1.15	1.71
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6828	519	499	588	606	623	592	602	600	562	567	542	528
	CDD18.3	3787	261	266	330	356	365	342	343	342	312	309	292	269
	CDH23.3	45082	2586	2966	4092	4791	4896	4489	4351	4257	3668	3289	3045	2653
	CDH26.7	19732	984	1167	1804	2464	2467	2145	1941	1847	1436	1195	1214	1067
Wind	WSAvg	2.4	2.1	2.5	2.8	2.7	2.5	2.7	2.7	2.7	2.1	1.8	2.1	2.4
Precipitation	PrecAvg	1244	26	22	59	77	151	150	116	103	224	243	62	11
	PrecMax	1647	117	89	214	237	318	375	267	263	385	411	194	42
	PrecMin	922	0	1	1	3	30	13	20	46	91	105	0	0
	PrecStd	185	30	22	49	55	79	84	69	54	75	79	53	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.5	33.6	34.2	35.6	35.3	34.9	34.2	34.0	33.9	33.9	34.2	34.2
		MCWB	23.9	24.7	26.2	27.2	27.6	27.1	26.8	26.5	26.6	26.3	25.3	24.9
	2%	DB	32.4	32.6	33.2	34.4	34.2	33.8	33.3	33.2	32.9	32.9	33.2	33.1
		MCWB	24.1	25.5	26.7	27.5	27.6	27.0	26.7	26.5	26.5	26.3	25.4	24.7
	5%	DB	31.8	32.0	32.7	33.8	33.7	33.1	32.6	32.5	32.1	32.2	32.6	32.2
		MCWB	24.3	25.6	26.7	27.5	27.5	26.9	26.6	26.4	26.4	26.2	25.3	24.5
	10%	DB	31.0	31.2	32.1	33.1	32.9	32.3	31.9	31.9	31.4	31.4	31.9	31.4
		MCWB	24.1	25.4	26.5	27.4	27.1	26.7	26.5	26.3	26.3	26.1	25.1	24.0
	0.4%	WB	27.2	27.8	28.6	29.2	29.4	28.6	28.6	28.4	28.2	28.2	28.1	27.3
		MCDB	30.8	31.3	32.0	33.2	33.5	32.1	31.5	31.3	31.2	31.5	31.6	31.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.6	27.2	27.8	28.6	28.7	28.2	28.0	27.7	27.7	27.6	27.3	26.5
		MCDB	30.2	30.7	31.7	32.6	32.7	31.6	31.1	30.8	30.7	31.0	31.1	30.7
	5%	WB	26.1	26.7	27.4	28.2	28.2	27.8	27.6	27.3	27.2	27.1	26.7	25.9
		MCDB	29.6	30.2	31.2	32.3	32.0	31.2	30.8	30.5	30.4	30.5	30.4	30.2
	10%	WB	25.6	26.3	27.0	27.8	27.8	27.4	27.2	26.9	26.8	26.7	26.1	25.3
		MCDB	29.0	29.8	30.8	31.8	31.5	30.9	30.5	30.3	30.0	29.9	29.7	29.3
Mean Daily Temperature Range	5% DB	MDBR	9.2	7.7	6.4	6.2	6.1	5.8	5.5	5.7	6.0	7.0	8.2	9.4
		MCDBR	9.9	8.5	7.2	6.8	6.9	6.5	6.1	6.4	6.7	7.9	9.2	10.0
		MCWBR	4.5	3.6	2.9	2.8	2.9	2.7	2.6	2.7	2.8	3.5	3.9	4.3
	5% WB	MCDBR	8.1	7.3	6.5	6.4	6.5	6.1	5.7	6.1	6.4	7.2	8.1	8.9
		MCWBR	4.1	3.3	3.0	2.9	3.1	2.9	2.8	2.9	3.1	3.5	4.0	4.3
Clear-Sky Solar Irradiance	taub	0.535	0.585	0.592	0.536	0.474	0.456	0.454	0.454	0.463	0.531	0.496	0.484	
	taud	2.016	1.902	1.929	2.107	2.332	2.399	2.399	2.386	2.344	2.114	2.196	2.195	
	Ebn at Noon	769	748	754	792	829	834	839	849	847	783	800	806	
	Edh at Noon	172	199	198	165	129	119	120	123	129	159	142	141	
All-Sky Solar Radiation	RadAvg	5.02	5.47	5.58	5.86	5.26	5.03	4.88	4.99	4.83	4.75	4.93	4.92	
	RadStd	0.33	0.28	0.43	0.39	0.36	0.27	0.29	0.33	0.28	0.33	0.38	0.26	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.16	N/A	N/A	N/A	+72	+72

Nomenclature: See separate page